

Lecture 10:-

Rules, Sum, Product, Conditional Probability

Rules of Probability:-

On a table, there are a total of 30 distinct books: 9 math books, 10 physics books and 11 chemistry books.

What is the probability of getting a book that is not a math book?

These events are ^{Physics} mutually exclusive".

$$\begin{aligned} P(\bar{M}) &= P(H) + P(C) \quad \text{no overlap!} \\ \text{not math} &= \frac{10}{30} + \frac{11}{30} \quad \text{chemistry} = \frac{21}{30} \end{aligned}$$

This is the "sum rule of probability". (Handling OR)

Generalize to include events that are not mutually exclusive:

A fair 20-sided dice is rolled.

What is the probability that the roll is an even number or prime number or both?

$$P(E \cup R) = P(E) + P(R) - P(E \cap R)$$

Double counted

E is even number

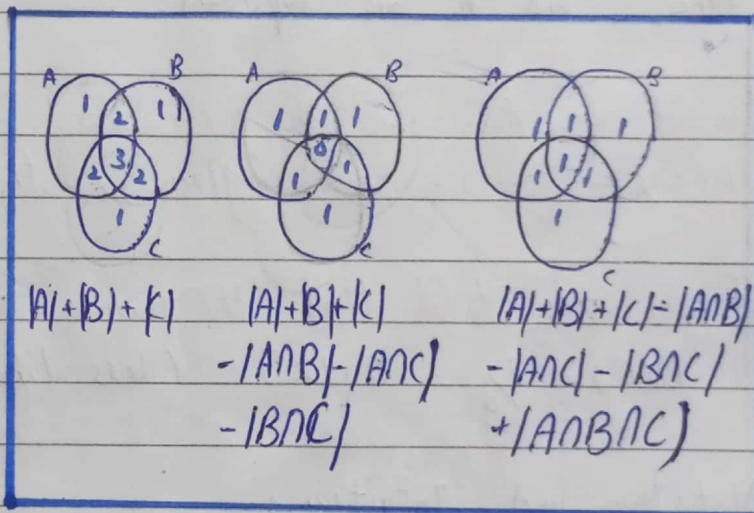
R is a prime number

$$= \frac{10}{20} + \frac{8}{20} - \frac{1}{20}$$

$$\Rightarrow P(E \cup R) = \frac{17}{20}$$

Inclusion Exclusion Principle:-

Again, the number in each subset represents how many times that subset is counted. Here, the overlap of all three subsets must be added back in due to it being removed after the second step.



It is a slight touch of this topic due your further research!!!

Product rule: (Handling AND)

Two coin flips: Probability of both being heads

$$P(HH) = P(H) * P(H)$$

Independent events = Knowing one has occurred doesn't change the probability of the other!

Mean if the Coin A is head the $P(H)$ for Coin B will always be 0.5 no matter what happen to Coin A

Conversely if $P(A) * P(B) = P(A \cap B)$
 then A and B are independent
 value

Example:- 20-sided dice is rolled. Probability that the number is even and prime. (the end shows that they have dependence)

Prove :- $P(E) = \frac{10}{20}$, $P(R) = \frac{8}{20}$ $P(E \cap R) = \frac{1}{20}$ $\therefore$ common value in both

because $P(E) \neq P(E) \cdot P(R)$
 $\frac{1}{20} \neq \frac{10}{20} \cdot \frac{8}{20}$
 $0.05 \neq 0.2$

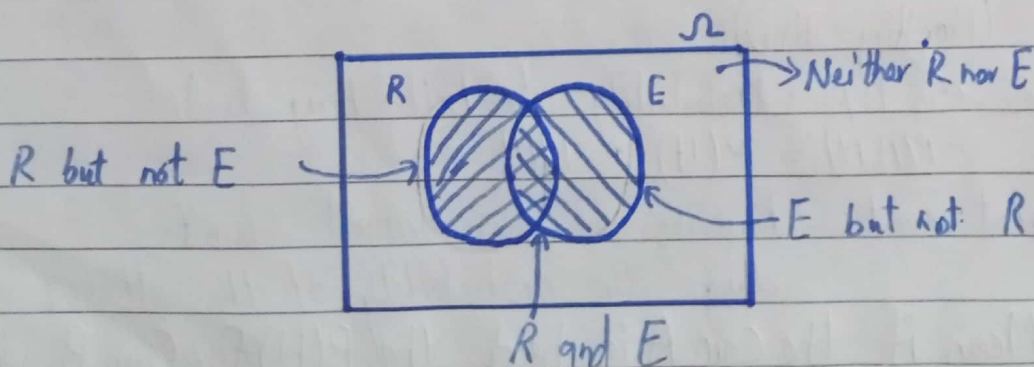
Hence, E and R are dependent.

So, if we know one, the probability of the other changes.
 $P(E) = \frac{10}{20} = 0.5$ "given no other information"

If you are told that the number was a prime
 $P(E) = \frac{1}{8} = 0.125$ (less likely now!)

★ Notation and Intuition:

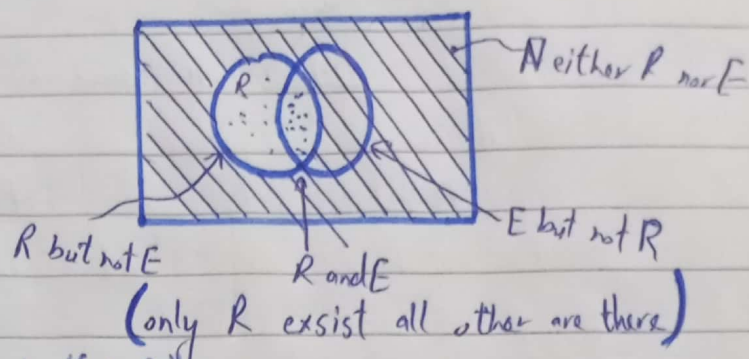
$P(E | R)$
 probability of event E given that this event is already known to have occurred!



Recall the axioms of probability
 The two axioms of probability
 - must lie in: $[0, 1]$

- Sum of all events must be 1

R is already known to have occurred! (given in question)



- R is our new "universe". There is not "not R".

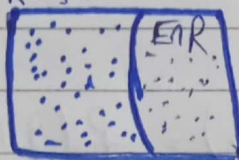
- R is definite - $P(R)$ has to be 1.

- But $P(R)$ was $8/20 = 0.4$.

- our math does not work!

- Rescale everything so that $P(R)$ becomes 1. $P(R)$ becomes $\frac{P(R)}{P(R)} = \frac{0.4}{0.4} = 1$

$$P(R|R) = 1$$



$$P(E|R) = \frac{1}{20} = 0.05$$

$$P(E|R) = \frac{P(E \cap R)}{P(R)} = \frac{0.05}{0.4} = 0.125$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Conditional Probability

Probability of A given that B has occurred $P(A|B) = \frac{P(A \cap B)}{P(B)}$ $\rightarrow$ Probability
 $P(B)$ $\rightarrow$ normalized so that axioms of probability still hold

$P(_ | B)$ ^{Imp} $\rightarrow$ You are transported to a ^{new} universe where 'B' is ^{already} true.
 Normalization is only necessary to make axioms of probability work

Otherwise:

0.125 is same as 0.05
 $\uparrow$ $\uparrow$
 $P(\Omega)$ $P(\Omega)$
 $\uparrow$ $\uparrow$
 1 0.4

In 0.125 , we say if universe is 1 then 0.125 is $P(E)$ which is equal to say in 0.05 if universe is of probability 0.4 then 0.05 is $P(E)$. but we know that the probability of universe is always "1" so first one is best way to say it.